Learning Engineering in Action

Agentic Simulation for Collaboration and Learning from Success and Failure

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Abstract

Learning engineering can be understood as a profession, a team practice, and an iterative process. We explore how these perspectives connect by executing role-based artificial intelligence (AI) agents on three fictional projects: workforce feedback, school-science reasoning, and public-library digital access. A learning-engineer agent orchestrates disciplinary and domain contributions through the LENS capability pipeline. A scenario agent introduces objections and unexpected conditions, while an append-only journal links discussion, decisions, and artifact revisions to runnable candidate products. The simulation makes proposed coordination responsibilities visible: whose concern changes a requirement, how disagreement is carried forward, and what happens when implementation contradicts a completed narrative. Replayable transcripts, evolving artifacts, inherited role contracts, and a reusable code framework provide materials for studying connection and collaboration. A source-grounded analysis examines privacy, reviewer conflict, and an actual implementation correction, alongside the additional cases' distinct access and participation demands. Successes and failures refer to observable project behavior, not demonstrated learning or human flourishing. The contribution is an inspectable method for examining team practice and competency definitions, with an evaluation agenda for its fidelity, teaching usefulness, and effects on actual collaboration.

Keywords: learning engineering; multidisciplinary teams; agentic simulation; design rationale; competency; evidence; human flourishing.

1. The work that a finished product hides

Some think of learning engineering as a profession, some as a team sport, and some as a process. Each perspective brings a different question into view. What should a learning engineer be able to do? How should a team bring its expertise together? How does the work proceed as a problem is understood, a solution is developed and evidence prompts revision? We use these perspectives together to explore competencies through the work they are intended to support.

In pursuit of clearer competency definitions, we examine the learning engineering iterative process, aided by advances in agentic AI technology. We execute real model calls in a fictional project: the problem context and team representation are simulated, while the agent contributions, revisions and generated code are actual outputs. We assign a learning-engineer role to orchestrate contributions from learning science, systems engineering, human-centered design, measurement, responsible AI, implementation and the relevant work domain. A scenario role introduces objections and unexpected events. This creates a representation of a team tackling a multifaceted problem and makes its proposed division of work available for discussion. The roles, skills and competency relationships are candidates to refine through professional and affected-person review.

We provide several simulation artifacts for readers to explore: a saved project replay, versioned designs and evidence plans, a runnable candidate product, and an explorer of reusable roles and proficiency targets. A reusable code framework supports new seeds, specialist contributions, scenario challenges and retained project records. These materials show how a proposed team arrangement operates across worked problems in different settings; they also let readers examine where the arrangement is incomplete. The replay includes three live-agent cases and a separately labeled scripted workforce rehearsal. [6-9]

A learning team receives a plausible request: help new team leads handle difficult feedback conversations. One response is to produce a course. Another is to investigate what the leads need to notice, decide and do, what prevents them from doing it, and whether instruction is the appropriate intervention. The second response requires a team to connect different kinds of expertise before it can justify a product.

A lead might lack a useful way to open a conversation. The colleague might have received incompatible instructions. The staffing arrangement might leave no time to clarify expectations. A manager might control both the working conditions and the interpretation of a worker's performance. These are different problems. A fluent rehearsal of a better opening does not resolve all of them.

The practical question is how a team works through such distinctions. Who challenges the initial explanation? Which assumptions become requirements? How does an objection change the design? What happens when the delivered code falls short of the design narrative? A polished final answer can conceal these decisions, including the decisions to narrow a claim, change the surrounding system or withhold a release.

We propose treating the evolving project record as a deliverable alongside the product. The record should let a reader follow a problem interpretation through specialist review, integration, implementation and correction. An agentic rehearsal is one way to construct a visible example of that work. Its usefulness as a representation of human teamwork remains an empirical question.

2. Grounding and contribution

The Institute of Electrical and Electronics Engineers (IEEE) International Consortium for Innovation and Collaboration in Learning Engineering (ICICLE) defines learning engineering through the learning sciences, human-centered design, engineering methodologies and data-informed decisions in support of learners and their development. The *Learning Engineering Toolkit* brings these concerns together with practical chapters on teaming, implementation and ethical decisions. These sources provide the professional context; they do not validate the simulation presented here. [1, 2]

The approach also draws on the distinction between building an artifact, demonstrating its use and evaluating the claims made for it in design science research. Sandoval's conjecture mapping provides a related

methodological discipline: make explicit how a design is expected to produce relevant processes and how those processes might lead to outcomes. We use these ideas to separate a visible interface behavior from a proposed learning mechanism and an eventual workplace benefit. This paper is a methods demonstration, not a completed design-based field study. [3, 4]

LENS, Learning Engineering for Next-Generation Systems, supplies the local capability frame: learner performance and system performance must be examined together, including the conditions that let a person act. The companion *Show Your Work* paper develops that argument and places human flourishing at the center of the intended benefit. [5] Here we examine how a represented team might coordinate the expertise needed to apply it. The eight stage names are this project's organizing vocabulary, not an adopted professional standard.

Our contribution is an inspectable connection among specialist contributions, challenged assumptions, decisions, executable artifacts, preserved failures, and bounded claims. We offer reusable role contracts and an evaluation agenda for whether that record helps people understand and practice collaboration. The value of the proposal depends on examination of those materials and their use, rather than on a claim of being the first to assign learning-engineering roles to agents.

3. What the team must coordinate

The learning-engineer role integrates the work. Integration means deciding what the problem is, what can reasonably be built, which questions require another discipline, and which decision should return to an earlier requirement. It includes preserving unresolved disagreement. It does not grant authority to speak for affected people or approve a real pilot.

The specialist roles divide attention rather than establish a hierarchy of credentials. Learning science examines the proposed practice, feedback and transfer mechanism. Systems engineering examines dependencies, competing causes and non-training changes. Human-centered design examines access, workload, agency and the perspectives still missing. Measurement examines what an observation could support and what rival explanations remain.

Responsible AI review examines delegation, privacy, consequential use and correction. Implementation review examines what the executable artifact actually does. Domain review connects the proposal to the supplied work context while naming where practitioner validation is needed. The scenario role introduces a plausible challenge to a particular assumption; it supplies hypothetical conditions, never invented observations.

Connection and collaboration require more than placing these contributions beside one another. A measurement concern must reach the design decision it constrains. A domain objection must be understandable to the implementation role. A requirement for participant control must survive the move from prose to code. The orchestrator's contribution is to integrate those relationships, explain the next decision, and preserve dissent that the integration does not resolve.

Together with the learning-engineer role, these are the nine named roles in the exemplar. They are design choices for this demonstration. A real team may combine functions in one person, distribute them across a larger group, or require other expertise. The relevant test is whether the necessary work is performed under appropriate conditions, not whether every role has a separate person or agent.

An agent's assigned disciplinary name establishes neither its expertise nor the independence of its judgment. The simulated roles share a provider and overlapping project context. Their agreement cannot be treated as expert consensus. In a human team, professional accountability, situated knowledge, relationships and power also affect what can be said and challenged. The role arrangement makes selected responsibilities visible without reproducing those conditions.

4. Method and evidence record

The exemplar begins with a fictional workforce seed: help new leads conduct difficult feedback conversations, consider a capability map and develop practice without employee surveillance. The supplied constraints exclude employee ranking and identifiable conversation recordings, require voluntary participation and evidence sharing, and require examination of staffing and workflow explanations. These constraints came from the project brief; they are not findings from workers or employers. [6]

The implementation coordinates role-specific, structured model calls through Understand, Map, Design, Build, Instrument, Experiment, Evaluate and Refine. Map and Experiment are the current public labels for stable storage keys `model` and `deploy`. Preserved journals retain their original stage prompts and agent wording; the display names do not imply that a new experiment was run. Each role receives the available project discussion and a bounded contribution request. Responses include advantages, disadvantages, tradeoffs, risks, human and system capability requirements, flourishing considerations, evidence references and dissent. The learning-engineer role synthesizes these contributions into an artifact or requests a return to an earlier stage. The Python runner schedules sequential calls and passes prior contributions to the next role; the learning-engineer agent returns integration decisions and artifact content. The runner validates the structured responses, journals them and assembles the artifacts. Specialist calls are bounded contributions and are instructed not to use tools or modify files directly. The roles are represented by actual model executions, rather than independent human participants or continuously operating workers.

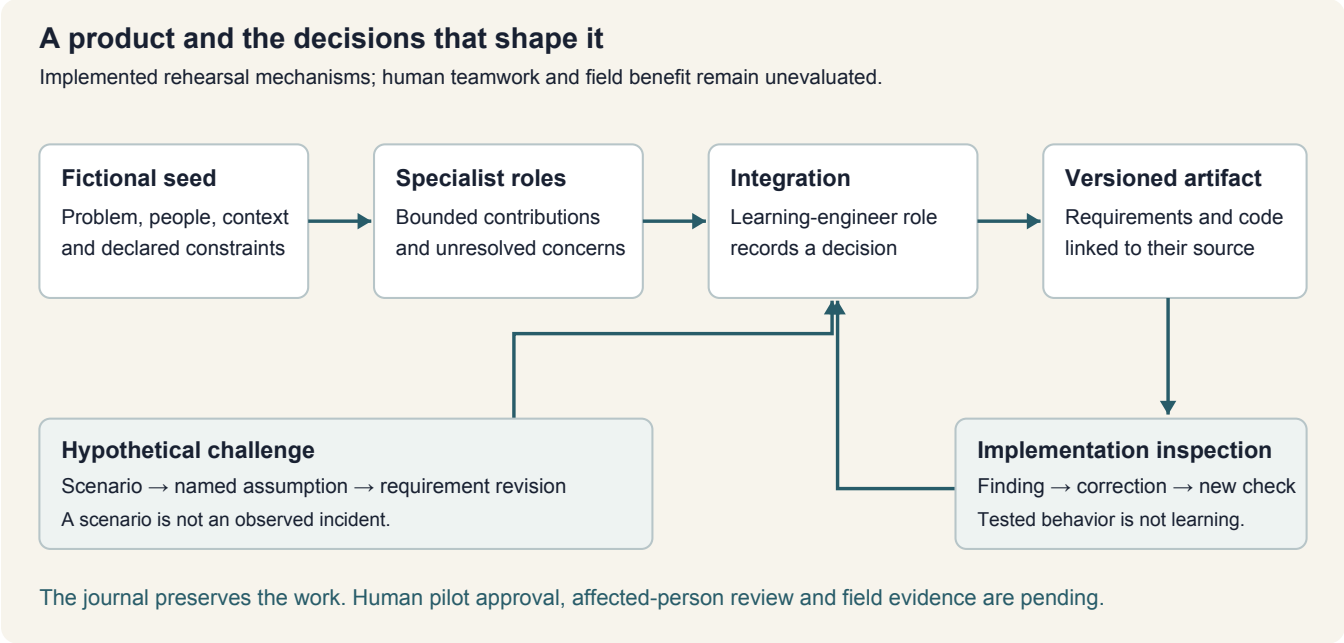


Figure 1. Proposed coordination method implemented in the fictional rehearsal. The arrows represent routing of work; they do not establish actual human-team behavior or field outcomes. Both challenged assumptions and observed implementation findings can return work to an earlier requirement.

The live run is actual agent execution on a simulated project. Its generated candidate is executable software. The fictional scenarios and proposed field evaluations are not observations of workers or workplaces.

A single preserved live-model journal, `20261001T224130Z-live-codex-d6a562`, is the evidence source for the detailed workforce episodes in this paper. It contains 99 events, 29 recorded model requests with returned structured responses, 14 artifact versions and two hypothetical scenario events. One corrective response was rejected and retained. These are inventory counts from the journal, not effect sizes, expert judgments or independent observations. Repeated completion markers reflect subsequent correction and review of the same project, not separate successful trials. A separately labeled scripted rehearsal is available but supplies no additional live-model evidence. [6]

The calls used the authenticated Codex command-line provider and the configured default model. The public metadata does not identify a fixed model version. The saved prompts, structured outputs and provider metadata permit inspection of this run; they do not guarantee that a future call will reproduce its content. Raw provider receipts and response caches remain local. No participant recordings or workplace observations are in the public record.

The journal links each event to its predecessor by hash and records artifact content hashes, revision parents and triggering decisions. The replay exposes only the discussion and artifacts available at the selected event. These checks detect changed records under the implemented verification rules. They do not sign an author's identity, verify the truth of a model's claim or prevent a new journal from being fabricated. The public text is returned deliberation and artifact content, not a representation of a model's private reasoning.

Our analysis is a retrospective reading of selected episodes. The episodes were chosen because they expose different coordination obligations: interaction design, governance of evidence and implementation correction. We did not preregister episode selection, independently code all contributions or estimate agreement among

analysts. The observations below describe linked events in this implementation; the implications for human teams are proposals.

Additional cases and selection rationale

Two additional live-agent projects use the same lifecycle with inherited domain specializations. They were selected to vary learners, setting and problem demands, not to represent a population or establish generality. The case explorer links their full records and candidate products; the case inventory reports run identifiers, event counts, journal hashes and pinned registry versions. [9]

The school-science case asks how middle-school learners might reason from conflicting fictional plant-growth observations. It distinguishes comparison from causal explanation and examines reading access, shared devices and classroom participation. The numbers in its invented evidence cards are story inputs, not experimental observations or results. Its domain profile adds classroom responsibilities without conferring teacher judgment or approval. Its first hypothetical challenge asks whether inserting suggested wording could overwrite a learner's qualified claim; the learning-engineer decision at event 29 adds replacement and recovery requirements.

The public-library case asks how adults might choose an acceptable next action for an unfamiliar fictional online service while retaining agency. It examines language access, assistance capacity and shared-device privacy. Online completion is not the sole objective: paper assistance, help-seeking, an undecided plan and stopping are legitimate possibilities. Its domain profile adds library responsibilities without access to real credentials or service records. Its first hypothetical challenge questions access to privacy controls during an interruption with magnified content; decision 29 specifies placement and focus requirements. The scenario supplies a condition to test, not an observed accessibility result.

Both cases preserve the agent contributions, hypothetical challenges and evolving artifacts alongside their generated candidates. They extend the software demonstration. Practitioner and affected-person review, analysis of differences among cases, and any actual learning study remain work to do. The three detailed episodes below are drawn from the workforce record; the cross-case discussion that follows is a descriptive comparison, not an independently coded analysis.

Case inventory and evidence boundaries

Case and source	Recorded execution	Distinct collaboration demand
Workforce feedback [6]	29 model calls; 14 artifact versions; 2 hypothetical challenges.	Connect privacy, staffing, review authority, and the actual implementation.
School science [9]	24 model calls; 11 artifact versions; 2 hypothetical challenges.	Preserve qualified claims and participation while exposing conflicting fictional evidence.
Public-library access [9]	24 model calls; 11 artifact versions; 2 hypothetical challenges.	Connect patron choice, shared-device control, and feasible assistance.

These counts come from each saved journal and can be checked against the case inventory. They describe software execution, not independent experts, human participants, or effect sizes. Host checks are separately recorded; final completion markers leave human pilot review pending.

5. Learning from success, failure, and revision

5.1 An interruption changes the privacy requirement

The early design treated session-only reflection, omitted analytics and absent stored history as important privacy boundaries. The scenario at event 27 introduced a different failure: a fictional lead leaves a shared workstation to cover a staffing gap, and a supervisor reads the reflection still on screen. The scenario challenged the assumption that avoiding persistence adequately protected interrupted reflection. No such workplace incident was observed.

The learning-engineer decision at event 29 requested a Design revision, producing revision two at event 30 and an explicit scenario-to-revision link at event 31. The new requirements addressed concealment, recovery and the interpretation of interruption. They also retained coverage as a work-system issue rather than labeling interruption a learner deficiency. [6]

This episode illustrates the connection between a technical boundary and the circumstances of use. The scenario did not merely add a caution to the final report; it changed an earlier artifact. The later candidate included hide, resume and discard controls. Their presence does not establish host privacy or prove that refusal is credible in a real workplace.

5.2 A reviewer conflict reopens the evidence protocol

The second scenario, at event 62, placed a hypothetical reviewer in charge of the staffing arrangement questioned by a response. The reviewer could interpret rescheduling as avoidance, while the participant would have to challenge their manager to correct the interpretation. The challenged assumption was that editable, descriptive review alone made evaluation meaningfully contestable.

The learning-engineer decision at event 64 requested a return to Instrument. The revised protocol at event 68, following decision 67, addressed reviewer conflicts, an independent correction route and withholding evaluative review when credible correction was unavailable. The revision retained a separate limit: reviewer independence cannot establish that the interpretation is correct. [6]

The coordination obligation here concerns the conditions that permit evidence to count. Adding a correction box to an interface cannot resolve unequal authority. A real team would need affected people and accountable owners to examine reviewer eligibility, consequences, resources and correction arrangements. The simulation identifies a requirement for that work; it does not perform or approve it.

5.3 Implementation review contradicts the completed narrative

The original Build artifact at event 38 contained a design draft but no executable HTML. The cycle nevertheless reached a completion marker. An implementation finding at event 81 reopened Build and required evaluation and refinement against actual source. This was an observed property of the software artifact, distinct from the two fictional scenarios.

The first corrective response returned code but omitted its triggering finding from its evidence references. Validation stopped at event 83. A later recorded explanation at event 97 identifies the prompt defect: the correction needed the finding's payload, not only its sequence number. The prompt was corrected, and decision 86 produced accepted Build revision two at event 87. The rejected output and earlier empty Build remain inspectable. [6]

The revised evaluation and refinement also narrowed the learning offer. The page organized reflection and supplied a fixed authored possibility; it did not analyze wording or deliver response-specific feedback. Decision 94 placed any proposed human facilitation in a separately reviewed pathway. At event 98, host checks recorded exercised reflection and correction propagation, labeled variation snapshots, concealment and focus behavior, resumption, discard, blank inputs and rescheduling. The final status remained awaiting a human pilot at event 99. [6]

This episode is particularly useful for teaching team practice because the narrative of completion was insufficient. Someone or something had to inspect the executable requirement, supply the finding to the next contributor and revise the interpretation of the delivered product. The record demonstrates that this correction occurred in the implementation. It does not show how reliably the process detects other defects.

5.4 The additional cases change what collaboration must protect

The school-science scenario at event 27 asks whether inserting a suggestion could overwrite a learner's qualified claim. Its Design revision responds with replacement and recovery requirements. The library scenario at the same event number asks whether privacy controls remain available during interruption with magnified content. Its revision responds with placement and focus requirements. The event number aligns with the same point in the lifecycle; the requirements differ because the valued work and conditions differ. [9]

This contrast makes a useful discussion task. A group can inspect the scenario, the integration decision, and the resulting artifact, then ask which specialist concern crossed into another role's work. In school science, assistance must preserve the learner's ability to qualify or dissent. In the library, a control's presence in the design is insufficient if it cannot be found at the moment of need. The source-specific host checks at event 81 record exercised behavior, while representative-user accessibility and field outcomes remain unverified. [9]

Across the cases, a requirement appearing in executable source is an observable implementation achievement. A missing requirement, rejected response, or unsupported completion claim is a failure worth preserving for instruction. Neither establishes a team's learning. Hypothetical scenarios identify plausible concerns; implementation findings identify observed properties of the artifact. Readers should keep these forms of evidence distinct when deciding what to revise.

6. What readers can inspect and learn from

Readers can replay the discussion and revisions, try the candidate product, explore role and competency contracts, and download the product and its project record. The reusable framework and running guide support local simulations with new seeds. The hosted site serves saved artifacts; it does not execute new model calls. Each case's artifact types expose different parts of its project. The additional cases diversify the represented problems; they supply no independent human-team observations. [6-9]

The implementation produced an executable candidate for fictional feedback practice and a portable handoff containing the candidate, requirements, current artifacts, full journal and a source-linked manifest. A reader can inspect the earlier drafts, the objections that changed them and the limits of the final product. The prototype is the work product; the evolving record explains why it takes that form. [6, 7]

The recorded technical checks support only the exercised behavior in their stated environment. Neither those checks nor the presence of a replay establishes accessibility for intended users, security of every generated artifact, appropriate workplace conditions or educational usefulness. In particular, the live run did not establish browser download completion; later packaging and publication checks verified archive contents and hashes separately. These are different observations and should remain distinct.

The record demonstrates coordination mechanisms in this software: role-specific contributions, explicit integration, scenario-linked requirement revision and retained correction. It does not establish that the mechanisms are sufficient for good teamwork. An extensive transcript can repeat a shared misconception. A hash-valid reference can point to a poor argument. An orchestrator can omit an important perspective, and a scenario role can generate objections that are too convenient or too weak.

Our proposed implication for human teams is that a project record can make important judgments available for review and instruction. An affected person should be able to ask why a requirement exists, who supplied its evidence and what happens if the assumption fails. Whether this particular record makes those questions easier to answer requires human evaluation.

Replay, artifacts, and a reusable framework

The landing page links all three cases to their replay, candidate and handoff. Replay exposes contributions, prompts, scenario-to-revision links and the artifacts available at each event. A separate Case demo tab shows the selected journal's latest executable Build, labeled with its revision and event; it follows the selected case, independently of replay position. Its source and manifest bind the candidate to the recorded artifact. A later requirement should not be read as something the team already knew at an earlier decision.

The role explorer shows how common responsibilities, disciplinary roles, domain specializations, and requested proficiency targets are resolved. The local code framework can run a new seed and preserve the resulting journal and handoff. The public website hosts saved execution and replay; browsing it does not launch new agents. Recorded real-model execution, scripted rehearsal, and newly authored teaching commentary are different materials and should retain their labels.

Why / teaching points for new learners

The companion Capability Pipeline presents five worked cases and a choose-your-own-adventure activity. Its current cycle is Understand, Map, Design, Build, Instrument, Experiment, Evaluate, and Refine. Authored teaching points explain why an activity matters, identify a decision cue, and ask a question in the case. Adventure choices add a rationale, tradeoff, and reflection after an outcome. Find-the-turn and predict-first modes retain their guess or reveal boundary rather than exposing the decision-specific explanation in advance. [10]

These teaching points are scaffolds, not additional agent observations or validated scoring rules. The adventure's draws remain fictional teaching assumptions, and its numerical outcome is not a measure of

decision quality. The live-agent journals retain their original wording and hashes. Guidance helps a reader examine the work while preserving what was actually recorded.

A possible learning activity is to pause at a challenged assumption, identify the missing perspective, and propose a justified next step before viewing the integration decision. Readers can then compare the requirement with the delivered artifact, explain where the connection held or failed, and consider a non-AI alternative. A new problem should test whether they can apply the same reasoning without the scaffold. These are proposed uses; their contribution to understanding or collaboration has not been measured.

7. Reusable roles and proficiency claims

The current implementation standardizes contribution contracts in registry 1.1.0 with competency framework 1.0.0. A common contributor profile supplies duties about evidence, uncertainty, dissent, correction, human and system capability, and flourishing. Specialist profiles add disciplinary responsibilities; workforce, school-science, and public-library profiles add context-specific requirements. Duties accumulate, while a child can replace a requested target for a named competency. Unknown parents, cyclic inheritance and unsupported target definitions are rejected. [8]

The associated competency vocabulary uses the five public LENS domains: Systems Analysis, Iterative Development, Human-System Collaboration, Test & Evaluation, and Sociotechnical Constraints. The proposed contribution levels are awareness, applied and integrative. These descriptors express the work requested from a simulation role. They are not adopted credentials, a certification crosswalk or evidence that a person or model possesses the stated proficiency.

A future claim of demonstrated proficiency would need a named subject, representative tasks, support conditions, criteria, reviewed observations, coverage and limitations. It should identify whether performance concerns a person, a team or a supported human-system arrangement. Inheritance can specify the expected contribution; it cannot supply that evidence or grant human approval authority.

The inherited profiles were added after the preserved workforce exemplar. That journal used its recorded fixed-role prompts. We do not retrospectively assign the later proficiency targets to it. The school-science and public-library cases pin their resolved profiles, framework versions and hashes in their first event. This distinction allows the code and roles to evolve while preserving what an earlier run actually used.

8. Human flourishing as a design obligation

The coupled capability claim concerns what a learner can do with a system that supplies usable information, support, authority, and correction. The workforce case connects judgment with staffing and review arrangements. School science connects interpretation of evidence with control over wording and participation. Library access connects a chosen next action with privacy and feasible assistance. A functional screen and a plausible learner response do not establish the complete arrangement's capability. [5, 6, 9]

Flourishing enters through concrete questions about agency, dignity, access, reciprocal usefulness and burden. Can a person decline practice or evidence sharing? Does the design confuse a staffing failure with a personal deficiency? Can a colleague challenge the lead's explanation? Who benefits from the intervention, and who supplies the time or emotional work it requires?

The design retains organizational repair, non-AI practice, rescheduling and stopping as legitimate possibilities. It excludes employee rankings and requires separate review before evaluative evidence is collected or used. These are declared design commitments and review requirements, not measured benefits. A visible button labeled decline cannot establish that declining has no consequences. A fictional exercise cannot establish that a colleague is treated more respectfully at work.

A human team would need to negotiate the intended benefit and unacceptable consequences with affected people, then investigate whether the proposed intervention and its surrounding conditions support them. Technical construction, educational value and the distribution of benefit require connected but different evidence. Keeping them separate helps identify what should be changed when a tool works as coded but the work system remains unsuitable.

9. An evaluation agenda for actual teams

The next study should begin with fidelity and usefulness of the representation, before attributing gains to agents. Learning engineers, domain practitioners and affected-person representatives could examine the same episodes and identify missing perspectives, implausible decisions, misleading authority assignments and requirements that the trace fails to carry into the product. Review should include disagreements and proposed changes to the simulation itself. This would supply human judgments about the representation, not learning-effect evidence.

A second study could test whether the record supports review. Reviewers could receive the same candidate product with or without its process record and answer predefined questions about assumptions, unresolved concerns and permissible release claims. Allocation, task order and the available time should be specified. Scorers could assess the responses against independently established artifact facts without knowing which record condition produced them. Outcomes might include detection of consequential gaps, appropriate narrowing of claims, trace accuracy, review burden and overlooked concerns. Completion speed alone would be insufficient. This comparison tests access to the record; it would not isolate the benefit of agents.

A third study could involve actual teams solving comparable problems with and without the proposed role and record protocol. The unit of analysis should include the team, its setting and support conditions. Depending on feasibility, allocation could be randomized or counterbalanced with explicit attention to carryover, prior expertise and resource differences. Proposed observations include whose expertise changes requirements, how disagreements are resolved or retained, whether defects are corrected, who bears added documentation work and whether reviewers can justify the next action. Candidate quality and team learning should be analyzed separately. Connection and collaboration also need observable criteria: whether one role's evidence changes another role's requirement, whether objections reach the accountable decision-maker, whether dissent remains inspectable, and whether a later contributor can explain the reason for a design choice. More messages or faster agreement would not, by themselves, demonstrate better collaboration.

For the workforce product, any field comparison would require credible voluntary participation, suitable access and coverage, an accountable owner, agreed evidence handling and stopping criteria. Retention and transfer would require observations across time and task variation, with attention to available support and opportunity to act. Prior experience, prompt familiarity, selection into participation, facilitation and changes to staffing are rival explanations. No sample size, performance threshold or effect estimate is supplied here; these require an explicitly justified study design.

10. Limitations and authorship

The detailed episode analysis is a retrospective interpretation of one selected fictional workforce project. Two additional fictional projects broaden the available demonstrations; their selection was purposive and their added transcripts have not undergone independent comparative analysis. It has no human-team observations, no untreated comparison, no independent disciplinary review and no participant outcomes. The model configuration is not pinned to a publicly identified version. The roles share a provider and project context; the same system helps generate and critique its proposals. Those dependencies limit any inference from apparent agreement or self-correction.

Scenarios were constructed to challenge assumptions, not sampled from an incident population. A missing-code finding demonstrates one discovered gap, not a measured defect-detection rate. The readable record may omit tacit decisions, cultural meaning, relationships and the consequences of speaking up. Keeping a transcript also creates work and potential exposure; a real team should collect only records justified by its purpose and evidence agreements.

The system and manuscript were developed with AI assistance under human direction. Model-generated contributions are labeled in the replay. This draft proposes a joint author argument and does not attribute completed research, technical work, stakeholder consultation or endorsement to either proposed author without confirmation. Before submission, both authors should review the claims, sources, interpretation, disclosures and contribution statement. The relevant standard is accountable human authorship, not treating generated text or a role label as an author.

11. Conclusion

Exploring learning engineering as a profession, a team sport and an iterative process connects competency questions to the work a team must perform. Teamwork becomes visible when a reader can follow a problem interpretation into a requirement, see an objection change that requirement and examine what the delivered artifact actually does. The live exemplars supply that connection through inspectable project records and runnable candidates. The method's strongest contribution is the availability of the work for critique, including a preserved implementation failure and a narrowed product claim.

The simulation offers a way to learn from success and failure without presenting either as a universal recipe. Its instructional value would come from examining the connections among expertise, assumptions, requirements, and consequences, then applying that reasoning to an unfamiliar problem. That value remains a hypothesis to investigate.

The next step is to test whether actual learning engineering teams and reviewers find this representation faithful and useful. A credible account should carry the product, the decisions that shaped it and the limits of the evidence together. That record can support a learning engineering team while leaving its professional judgment, relationships and accountability with people.

References and inspectable materials

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